Benjamin Irving

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Available: June 2027

Education

UC Santa Cruz, Baskin School of Engineering 2026-present
PhD in Computer Science

Northeastern University, Khoury College of Computer Sciences 2020-2024
Bachelor of Science in Computer Sciences, minor in Physics GPA: 3.7/4.0 | Dean's List

Relevant Courses: Group Theory | Reinforcement Learning | Machine Learning and Data Mining 2 | Robotic Science and Systems | Computer Systems | Physics for Engineering 1 | Software Engineering

Industry Experience

Software Engineering Intern, Vecml September 2024 – September 2026
- Designed and implemented our Agentic System (coding agent, planning agent, ppt agent, etc.)
- Implemented our code-execution backend, which uses docker and berkeley packet filters (BPFs) along with linux system calls. Compatible with CUDA, and our in-house AutoML backend
- Contributed to our Retrieval Augmented Generation (RAG) system, implementing fast Table RAG

Machine Learning SDE Intern, Amazon Web Services, Annapurna Labs June 2022 – Aug 2022
Cupertino, California
- Implemented optimized Transformer encoder models (BEiT, ViT) using Neuron-specific ops for the Amazon Tranium chips
- Created extendable repository for AWS-Neuron team, benchmarking throughput, latency, and cost

Quality Engineer Intern, Optum (UnitedHealth Group) Jan 2022 - June 2022
Boston, Massachusetts
- Automated test suite using Java, Maven, and Selenium, reducing test workload from 96 to 4 hours
- Implemented test automation for Optum Financial (1.8M customers, $200M annual claims)

Publications

1. Irving, B., & Schoene, A.M. (2024). "MEANT: Multimodal Encoder for Antecedent Information." In *Proceedings of EMNLP 2024*, pages 8579-8600.
2. Zevallos, R. J., Ortega, J. E., & Irving, B. (2024). "Related Work Is All You Need." In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, pages 13874-13878.
3. Schoene, A.M., Garverich, S., Ibrahim, I., et al. (2024). "Automatically extracting social determinants of health for suicide: a narrative literature review." *npj Mental Health Research*, 3(51). doi:10.1038/s44184-024-00087-6

Research Experience

Adversarial Research, Khoury College at Northeastern University May 2025 – Present
- Attacking VLAs. Working with traditional RL methods and Diffusion.

Research Assistant, Institute for Experiential AI Oct 2023 – Dec 2024
- Training large language models on mental-health data using multiple GPU nodes
- Implementing extensible NER library supporting nested-NER capabilities
- Building infrastructure with PyTorch, Cython, Triton, OpenMPI, and HuggingFace libraries

Undergraduate Researcher, JSALT NLP Workshop, Johns Hopkins University June 2023 – Aug 2023
Le Mans, France
- Developed large-scale graph embedding procedures for 200M+ nodes using randomized SVD and Chebyshev iterations with C, Python and Cython
- Reduced runtime by 24% and memory requirements by 25% (200GB) through optimized implementation
- Project: `github.com/kwchurch/JSALT_Better_Together`

Reinforcement Learning Research April 2024 – June 2024

- Trained deep reinforcement learning models to complete tasks with a robotic arm, working with equivariance and Proximal-Policy Optimization (PPO) to improve sample efficiency.